

AccessibleShip: Exploring Packaging and Shipping Barriers for Visually Impaired Users in C2C E-Commerce

Kunal Gupta
Empathic Computing Laboratory
The University of Auckland
Grafton, Auckland, New Zealand
kunal.gupta@auckland.ac.nz

Ken Takaki
The University of Tokyo
Bunkyo-ku, Tokyo, Japan
takaki@akg.t.u-tokyo.ac.jp

Ayaka Tsutsui
Digital Nature Group
University of Tsukuba
Tsukuba, Japan
Mercari R4D
Tokyo, Japan
ayakatsutsui1128@digitalnature.slis.tsukuba.ac.jp

Kentaro Teramoto
Mercari R4D
Tokyo, Japan
zigen@mercari.com

Bektur Ryskeldiev
Mercari R4D
Tokyo, Japan
Research and Development Center for
Digital Nature
University of Tsukuba
Tsukuba, Japan
ryskeldiev.b@gmail.com



Figure 1: Real-world examples illustrating the challenges faced by visually impaired users during the C2C selling process: a) difficulties filling out handwritten shipping slips, b) packaging items securely with oversized boxes, and c) navigating cluttered app interfaces on C2C platforms

Abstract

People with Visual Impairment (PVI) face numerous challenges when engaging in Customer-to-Customer (C2C) e-commerce, particularly in tasks like listing, packaging, shipping, and post-sale interactions. This case study investigates the accessibility barriers that prevent visually impaired users from independently managing the selling process on C2C platforms. Through semi-structured interviews with eight participants with low vision, we identified challenges, such as reliance on assistance, inaccessible platform interfaces, and difficulties in offline tasks. The study highlights

the need for AI-powered assistive tools, accessible design improvements, and integrated post-sale management systems. Our findings contribute actionable insights for inclusive design practices, aiming to foster independence and satisfaction for visually impaired users in digital marketplaces.

CCS Concepts

• Applied computing → Online shopping; • Human-centered computing → Empirical studies in accessibility; Accessibility design and evaluation methods.

Keywords

accessibility, e-commerce, packaging, online shopping, user studies

ACM Reference Format:

Kunal Gupta, Ken Takaki, Ayaka Tsutsui, Kentaro Teramoto, and Bektur Ryskeldiev. 2025. AccessibleShip: Exploring Packaging and Shipping Barriers for Visually Impaired Users in C2C E-Commerce. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*,

April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 8 pages.
<https://doi.org/10.1145/3706599.3706665>

1 Introduction

Online shopping offers convenience and autonomy, particularly benefiting People with Visual Impairments (PVI). E-commerce platforms simplify tasks like browsing, selecting, and purchasing products, which can be challenging in physical stores [12]. Additionally, customer-to-customer (C2C) platforms empower PVI by enabling them to sell unused goods, fostering financial independence and helping to declutter their homes. However, while accessibility improvements have enhanced the buying experience, significant barriers persist in the selling process on C2C platforms.

C2C platforms differ from traditional e-commerce by requiring both online and offline tasks, many of which lack accessibility. Overloaded user interfaces, limited screen reader support, and inadequate assistive technology integration make it difficult for PVI to navigate, write descriptions, and manage listings [18]. The post-sale experience, including communication, shipping, tracking, and packaging quality, plays a key role in customer satisfaction and repurchase intentions [11]. While external entities manage elements like logistics, sellers control packaging quality. Self-packaging in C2C e-commerce poses difficulties for those with impairments, who may lack the skills or physical ability to ensure consistent, high-quality packaging. Poor packaging can lead to dissatisfaction, negative reviews, or returns. Routine tasks manageable for sighted users become major obstacles for PVI, with research showing that up to 70% of PVI refrain from selling on C2C platforms due to these compounded challenges [14].

This exclusion from the secondhand economy highlights a critical gap in accessibility, limiting financial independence and full participation in digital economies for PVI. While efforts to improve accessibility in e-commerce have focused primarily on the buying experience, the selling process remains underserved. Offline tasks like photographing items, packaging, and shipping are complex and largely inaccessible for PVI. Given the growing reliance on C2C platforms for sustainable living and financial empowerment, addressing these accessibility barriers is essential. This study explores factors contributing to a positive, accessible selling experience for PVI on C2C platforms. We investigate how assistive technologies and inclusive design can enhance independence throughout the selling process, including post-sale stages like packaging, shipping, and tracking. In this pilot study, we address two research questions:

RQ1: What are the accessibility barriers that prevent visually impaired users from independently engaging in the selling process on C2C platforms, and how do these barriers affect buyer satisfaction?

RQ2: How can assistive technologies and inclusive design innovations enable visually impaired users to independently manage the selling process, from listing to post-sale interactions?

To answer these questions, we conducted a study consisting of a questionnaire and semi-structured interviews. Eight participants with visual impairments were recruited, and a thematic analysis using a grounded theory approach was employed to interpret the data. The following sections present the findings, lessons learned, and broader implications of our results.

2 Background

2.1 Introduction to Online Shopping and Accessibility Tools for PVI

Online shopping is crucial for People with Visual Impairments (PVI), offering access to products and services from home, which is essential due to mobility challenges [20, 21, 25]. Studies show that 93% of blind individuals and 96% of those with low vision shop online at least occasionally [14]. HCI research has made significant strides in improving website and app accessibility, enhancing the shopping experience for PVI. Advances like screen readers, voice assistants, and browser extensions have improved navigation and e-commerce site accessibility [17, 23]. AI-based tools now help structure online shopping platforms [19], while voice assistants optimize web search for PVI [1, 2, 6, 22, 24], collectively boosting their autonomy in the online domain.

Some PVI now sell items on online marketplaces, leveraging tools for tasks like photographing and listing products. However, significant challenges remain in packaging, shipping, and tracking. AI has advanced automatic content description generation [4, 8] and improved photo quality [7], offering PVI more opportunities as sellers. Despite these improvements, offline processes—especially packaging and shipping—remain a barrier to fully independent selling [18]. This study investigates solutions to empower PVI in managing these offline tasks, aiming to improve accessibility and participation for PVI in online marketplaces.

2.2 Challenges in Offline Aspects of Online Shopping for PVI

While online shopping offers PVI convenience, many offline challenges remain unresolved. Technological efforts have aimed to improve offline shopping independence, such as navigation systems integrated into canes for supermarket shopping [13], barcode scanners like ShopTalk for product identification [15], and TineTra, which uses RFID readers and GPRS for product information retrieval [10]. Recent research has explored conversational interactions using large language models (LLM) to help PVI navigate shopping malls [9], and computer vision with augmented reality (AR) for assisting with parcel lockers [5].

However, these technologies do not fully address challenges related to offline sales processes, particularly product packaging and shipping, where PVIs often depend on friends, family, or store clerks for assistance [26]. Although PVIs express a strong desire for greater independence in these tasks, current research has not sufficiently explored these offline aspects of selling [18]. Our study seeks to evaluate these specific barriers and propose solutions to enhance the selling experience for PVIs, ultimately aiming to create a more inclusive e-commerce environment.

3 Methodology

We adopted a qualitative research approach to explore the challenges faced by PVI in selling on C2C platforms and how assistive technologies might help. We used semi-structured interviews and user journey mapping, drawing from established HCI methodologies in previous accessibility and e-commerce studies [18, 20].

Participant: Following institutional review board approval, we recruited eight participants with low vision (four males, four females), aged 20–54 (Mean = 26.5, SD = 9.67), through snowball sampling. A screening questionnaire gathered demographic details, type and duration of impairment, experience with assistive technologies, and prior involvement in online sales (see the appendix for further details). Four participants had congenital impairments, while others had been affected for 10–22 years. Participants regularly used assistive tools such as text magnification, color filters, screen readers, and monocular aids. All participants had an online shopping experience, though only six had used C2C platforms like Mercari¹ and Rakuma². Of those, four (P1, P2, P3, P7) had experience selling; those without cited concerns about verifying product images, screen reader reliance, and platform usability.

Semi-Structured Interviews: We conducted semi-structured interviews based on methods from Ryskeldiev et al. [18] and other accessibility studies. The questions focused on participants' experiences with C2C platforms, covering tasks like item listing, packaging, and shipping. The interviews followed the complete user journey, addressing both online (e.g., listing, navigation) and offline tasks (e.g., packaging, shipping), as well as post-sale activities like tracking and feedback management. Each 60-minute interview via Zoom³ allowed participants to discuss their challenges and strategies. Interviews conducted in Japanese were transcribed into English using Whisper⁴, and the translations were reviewed to ensure accuracy and nuance. Participants were compensated with 10,000 Japanese yen.

3.1 Data Analysis: Thematic Analysis and Journey Mapping

We combined thematic analysis with user journey mapping to capture PVI's experiences in C2C e-commerce, using these methods to identify patterns and visualize user interactions, particularly in accessibility contexts.

Thematic Analysis: To streamline initial theme exploration, GPT-4o [16] was used to assist in identifying potential patterns and recurring challenges in the data (similarly to [3]). The AI-generated insights acted as a preliminary step, highlighting potential themes such as usability issues and dependence on assistance. However, two researchers manually reviewed, refined, and validated all themes through independent coding. Manual coding allowed for deeper insights, such as uncovering issues such as small text sizes or emotional responses that the AI had only broadly identified. Discrepancies between AI outputs and manual analysis were resolved collaboratively, ensuring a thorough interpretation. The key themes were then finalized and presented in the results.

User Journey Mapping: User journey mapping was used alongside thematic analysis to visualize participants' experiences throughout the C2C selling process. This method mapped out key stages—such as listing items, packaging, shipping, and post-sale

interactions—highlighting pain points like difficulty selecting packaging and managing post-sale feedback. It also captured participants' emotional responses, such as frustration and anxiety, helping contextualize their experiences. This approach further emphasized opportunities for design improvements, reinforcing the thematic findings and offering a holistic view of the challenges and aspirations of PVI in C2C platforms.

4 Findings

Our analysis identified several key themes regarding the challenges and experiences of PVI in C2C e-commerce. These themes span various stages of the selling process, revealing barriers related to platform usability, offline tasks such as packaging and shipping, and the need for greater independence. Below, we present the findings based on these recurring themes.

4.1 Key Themes

Dependence on Assistance. A prominent theme was the reliance on assistance from family members or friends to complete various tasks associated with selling items on C2C platforms. Several participants (P1, P2, and P4) expressed that visually demanding tasks, such as packaging, taking photos, and navigating the platforms' user interface (UI), could not be performed independently. For example, P1 stated: *"I always ask my parents for help with packaging because I can't manage the small screens."* This dependency on others created a barrier to their autonomy and contributed to their hesitancy in engaging more actively in C2C platforms.

Challenges Due to Visual Impairments. Another prominent theme was the difficulty participants faced in performing tasks that required precise visual input, such as reading small text, taking clear product photos, or verifying item details. Participants frequently mentioned their struggles with using small smartphone screens and apps that lacked adequate accessibility features. P3 shared, *"Even with the best camera, it's still hard to see small scratches on the items."* This issue was compounded by challenges in framing product photos properly, as noted by P4: *"I struggle to get the angle right in photos; I can't tell if it's framed properly."* The inability to complete these visual tasks independently meant that participants had to rely on others or face dissatisfaction from buyers if item descriptions or photos were inaccurate.

Packaging Practices and Challenges. Packaging items emerged as a critical pain point for many participants. Deciding on the appropriate packaging materials, particularly for fragile or large items, was cited as a challenge, especially when participants were unsure if their packaging methods were secure. P2 expressed concern over the safety of items during shipping: *"I'm always worried that the items might move inside the box and get damaged, so I use extra bubble wrap."* These challenges highlight the need for packaging solutions that are both accessible and tailored to the needs of visually impaired users, reducing their reliance on others and improving their confidence in packaging fragile items securely.

Accessibility Issues with C2C Platforms. The usability of C2C platforms was another key issue identified by participants. Many apps used for listing and selling items were described as having small

¹<https://jp.mercari.com/>

²<https://fril.jp/>

³<https://zoom.us/>

⁴<https://openai.com/index/whisper/>

fonts, cluttered layouts, and poor contrast, all of which posed significant barriers for visually impaired users. P1 described the interface of a C2C marketplace app as particularly difficult to navigate: *“The app feels cramped, and the images are too small to properly see.”* Similarly, P2 mentioned that *“increasing the text size in the app led to overlap in the interface”*, further complicating navigation. This theme underscores the need for C2C platforms to adopt more inclusive design practices prioritising accessibility, ensuring visually impaired users can complete the selling process independently.

Shipping Preferences and Challenges. Shipping posed additional challenges for participants, particularly when filling out manual shipping slips and managing delivery logistics. Several participants preferred online shipping label generation and home pickup services, but not all were aware of these options. P3, P5 and P8 stated that although they knew about online label printing, they had not used the service. Additionally, P4 highlighted the importance of helpful staff in the shipping process: *“I prefer Yamato (a domestic shipping company) because the staff always help me pack carefully.”* These findings suggest a need for more user-friendly shipping options that cater to the accessibility needs of visually impaired sellers, reducing the barriers they face in the post-sale phase of the selling process.

Post-Sale Interaction and Feedback Management. After shipping the item, visually impaired sellers faced further difficulties managing post-sale tasks, such as tracking deliveries and responding to buyer feedback. P4 and P6 mentioned that reading delivery tracking information was challenging due to small text sizes, with P1 and P4 were disengaged from interacting with post-sale feedback due to the platform’s complexity or reliance on others to interpret buyer comments. P1 shared, *“I don’t really look at the feedback, my parents check for me if there’s a problem.”* These findings emphasize the need for C2C platforms to simplify the post-sale experience, ensuring that visually impaired sellers can easily manage feedback and tracking.

Desire for Independence. Finally, a strong desire for greater independence emerged as a central theme across all participants. Many expressed frustration with their reliance on others for tasks such as packaging, shipping, and interacting with buyers. However, participants also believed that with the right tools and accessible designs, they could handle the selling process independently. P4 said, *“If I had access to all the information I need through my phone, I could do more on my own,”* reflecting a widespread sentiment. This desire for independence underscores the need for designing assistive technologies and accessible platform features that empower visually impaired users to navigate the selling process autonomously.

4.2 User Journey Mapping Results

To gain deeper insights into the selling process for visually impaired individuals on C2C platforms, we constructed a user journey map based on the participants’ experiences. The map outlines each stage of the selling journey, from the initial decision to sell through to post-sale interactions, while highlighting key pain points, challenges, and reflections at each stage.

Stage 1: Deciding to Sell. The journey begins with participants deciding whether or not to sell items. Their perception of the complexity of handling visually demanding tasks often influenced this decision. Some participants hesitated to sell items due to concerns over managing tasks such as taking photographs and packaging items. For example, P4 expressed reluctance to sell, stating: *“I don’t know how to sell online, and taking photos for shipment seems difficult.”* Similarly, P1 and P3 echoed concerns about the complexity of the process, which deterred them from engaging more frequently in online selling. P5 highlighted the challenge of determining an appropriate selling price due to visual fatigue caused by navigating listings on C2C platforms. P5 explained: *“I think there are a lot of products with different prices, but when I compare them, I have to go here and there, and my eyes get tired.”*

Stage 2: Listing the Item. Once the decision to sell was made, participants listed the item. This stage involved photographing the item, writing a description, and setting a price. A common challenge for participants was seeing small details on their listed items. For example, P7 mentioned: *“A couple of times when I sold clothes, there were stains that I couldn’t see, and the people who bought them found them, so communication didn’t go well.”* Participants also faced issues with photographing items, as P4 had trouble framing the items correctly. In addition to these challenges, some participants found the app interfaces cumbersome, with cluttered layouts and small text, making completing the listing process efficiently difficult.

Stage 3: Choosing Packaging Materials. Selecting appropriate packaging materials presented additional challenges for participants. P1 and P3 struggled to find materials suitable for larger or more fragile items. P1 remarked: *“I struggle to find materials for larger items like monitors.”* Concerns over packaging security were also prevalent, with participants worrying about whether their items would be sufficiently protected. P2 shared: *“I’m always worried that the items might move inside the box and get damaged.”* Additionally, P5 faced difficulties in physically accessing packaging materials, as they could not travel easily to stores that sold such items.

Stage 4: Packaging the Item. For the packaging itself, many participants relied heavily on others, such as family members or friends, to assist with this task. P1 noted: *“I rely on my parents to help with packaging due to the small screen.”* Similarly, P4 expressed discomfort with their packaging skills, stating: *“It’s embarrassing when the packaging doesn’t look neat, so I avoid wrapping.”* For participants who packaged their own items, the process often took longer than expected, as they had to carefully check the item’s surface for irregularities. P7 described using an iPhone camera’s magnifier to navigate this task: *“I use the iPhone camera’s magnifier to ensure neat packaging. I primarily use it to confirm whether the thickness is under 3 centimetres by placing a ruler and checking the markings precisely.”* Such efforts reflect the additional cognitive and physical load placed on participants due to their visual impairments, further highlighting the importance of accessible packaging tools or technologies that can simplify these tasks.

Stage 5: Shipping the Item. Shipping proved to be a significant hurdle for participants, particularly when filling out hand-written shipping slips. P1, P3, and P8 expressed frustration with the manual process due to their visual impairments. P1 shared, *“It’s hard for me to see the*

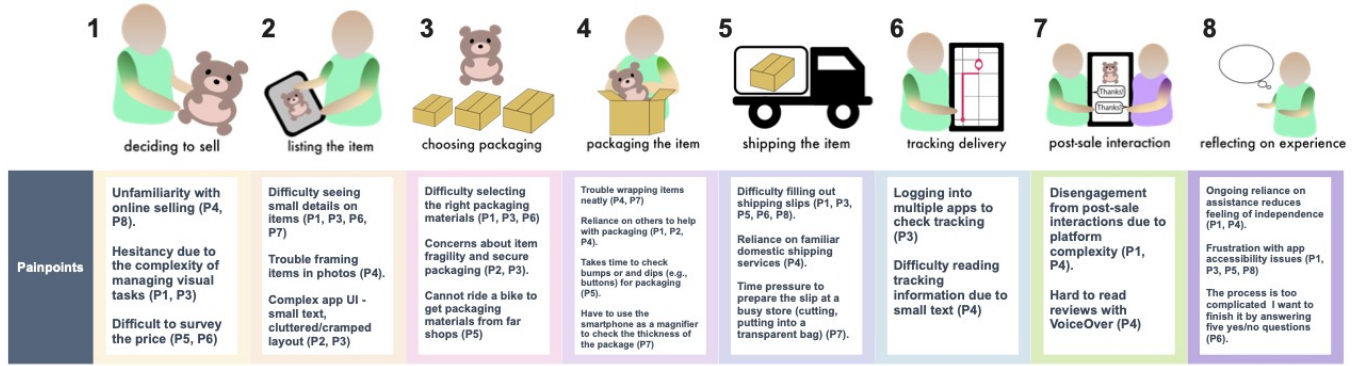


Figure 2: User Journey for visually impaired sellers on C2C platforms. The journey consists of seven stages: (a) Deciding to Sell, (b) Listing the Item, (c) Choosing Packaging, (d) Packaging the Item, (e) Shipping the Item, (f) Tracking Delivery, (g) Post-Sale Interaction, and (h) Reflecting on the Experience.

lines and fill out the details correctly.” Some participants preferred digital solutions that allowed them to input shipping details online and print labels at home. However, not all were aware of these options. P3 mentioned, “I didn’t know that I could do everything online, so I’ve been filling out the slips by hand.” This lack of awareness about digital alternatives further complicated the shipping process for many participants. For those who preferred more hands-on support, arranging home deliveries or visiting the closest convenience store were common choices. Some participants used their parents’ usual shipping services, while others, such as P4, chose delivery services offering extra assistance. “I use domestic shipping company, because the staff help pack the items and handle everything for me” (P4). This preference for supportive services underscored the need for accessible shipping options that reduce the burden on visually impaired users. The reliance on family members for shipping tasks was also notable. Several participants, including P2 and P5, mentioned that their parents often managed the shipping process on their behalf. This not only highlighted the accessibility challenges associated with shipping but also reinforced the broader theme of dependence on others throughout the C2C selling journey.

Stage 6: Tracking Delivery. Once the item was shipped, participants encountered additional challenges in tracking delivery. Tracking information, often presented in small text, proved difficult for visually impaired participants to access. For instance, P6 expressed frustration with this aspect of the process, stating: “I cannot see, so I’m not interested in seeing tracking status. It’s an enjoyment for people who can see.” Similarly, P4 mentioned struggling to read the small text used for tracking updates in the app.

Stage 7: Post-Sale Interaction. Managing post-sale interactions, such as responding to buyer feedback or checking reviews, presented another barrier. P1 and P4 felt disengaged from these tasks due to the complexity of the platform interface. P1 noted: “My parents contact buyers once the product is delivered. In the beginning, I tried to read the comments, but the letters were too small for me.” This reliance on family members for post-sale interactions underscored the inaccessibility of certain platform features, such as reading

buyer feedback. While some participants were indifferent to post-sale interactions due to their inability to access this information, others were motivated to ensure buyer satisfaction. These participants made extra efforts to take better photos or ensure the quality of their listings, as they were concerned about receiving negative reviews. For example, P7 shared that they wished for a system to help identify any dirt on items before shipping: “... if someone says the item looks dirty or the packaging is sloppy...” A unique strategy was also shared by P2, whose mother often added extra promotional items with the order to boost buyer satisfaction: “My mother sometimes adds extra promotional items for customers, especially for idol merchandise, to please them.” This proactive approach was an effort to avoid negative reviews and create a positive post-sale experience for buyers. While not all participants were heavily invested in post-sale interactions, those who desired clearer, more accessible systems could help them manage feedback and ensure their buyers’ satisfaction.

Stage 8: Reflecting on the Experience. Finally, participants reflected on their overall experience with C2C selling. P1, P3, P5, P6, and P8 expressed frustration with the accessibility barriers they encountered throughout the process. P1 and P4 noted that they felt less independent because they relied on others to complete tasks. P6 pointed out the financial strain caused by needing assistance, explaining: “If I ask a friend to display an item and I give them 10% of the sale as a reward, it doesn’t make sense to put it on a C2C market as the profit is not very big anyway. It would be great if AI could do that part.”

5 Discussion

Our study explored the accessibility barriers and potential design innovations needed to support visually impaired users in independently managing the selling process on C2C platforms. Below, we discuss the findings in relation to our research questions.

RQ1: What are the accessibility barriers that prevent visually impaired users from independently engaging in the selling process on C2C platforms, and how do these barriers affect user satisfaction?

Lack of Awareness of the Shipping Process and Existing Digital Tools. One of the primary barriers identified was a lack of awareness about the shipping process and the existence of accessible digital tools. For participants like P4, the complexity of managing visual tasks like taking photographs and preparing items for shipment discouraged them from selling: *“I don’t know how to sell online, and taking photos of items for shipment is difficult, so I have never tried.”* Many participants like P1, P3, and P8, who are unfamiliar with C2C platforms, kept themselves away from the selling process because they did not know how complicated the process would be, and it seemed to be difficult for them. Furthermore, filling out handwritten shipping slips was repeatedly mentioned as a barrier. Although some participants were aware of digital alternatives, like completing the slips online, they still had no experience using them. Furthermore, one participant was completely unaware of such digital solutions. Consequently, this lack of awareness of existing digital tools and preconceptions about the complexity of selling on C2C platforms prevent potential visually impaired (PVI) sellers from engaging in the selling process.

Dependence on Assistance for Offline Tasks. The selling process on C2C platforms combines online and offline tasks, often presenting compounded challenges for visually impaired users. Many participants, particularly those unfamiliar with C2C platforms, expressed frustration and disappointment with their inability to navigate these tasks independently and relied on family or friends’ assistance. For example, P1 commented *“I rely on my parents to help with packaging due to the small screen.”* This experience discouraged them from actively engaging as sellers, despite their interest in participating in the secondhand economy. This reliance significantly undermined their sense of autonomy and contributed to feelings of frustration. Participants expressed a desire to handle these tasks independently but felt limited by the physical demands of packaging and the lack of accessible tools to guide them through the process. The inability to complete these critical offline tasks without assistance is a substantial barrier to full participation in C2C platforms. This, in turn, affects user satisfaction, as the participants feel they cannot manage the selling process on their own terms.

Poor Post-Sale Interaction Interfaces. As we observed in the thematic analysis and user journey mapping, many participants were disengaged from post-sale interactions, such as responding to buyer feedback or checking reviews, because of the small text size and the complexity of the platform interface. However, two participants were strongly motivated to engage in the process by seeking help from family members to ensure buyer satisfaction. This disengagement from post-sale interactions due to poor interface design diminishes users’ sense of independence and limits their ability to engage fully in the C2C process. These barriers to effective post-sale engagement affect user satisfaction and prevent visually impaired sellers from building trust and maintaining relationships with buyers.

RQ2: How can assistive technologies and inclusive design innovations enable visually impaired users to independently manage the selling process, from listing to post-sale interactions?

The findings from our study highlight a clear need for assistive technologies and inclusive design innovations that can support visually impaired users in independently managing the selling process on C2C platforms. Our findings reveal several opportunities for improving platform accessibility, usability, and autonomy at different user journey stages, from item listing to post-sale engagement.

AI-Assisted Tools for Packaging and Shipping. A key finding from the interviews was the difficulty visually impaired users face in selecting appropriate packaging materials and ensuring the safety of items during shipping. Participants like P1 and P3 struggled to find packaging for larger items and expressed concerns about whether their fragile items were well-protected. There is a clear opportunity to develop AI-powered tools to guide users through packaging, helping them select the right materials based on item type and size. For instance, an AI system could provide step-by-step verbal instructions or tactile feedback to ensure an item is securely packed. P3 suggested, *“If I could get information through my smartphone, I think I could handle it...”*, indicating that AI-driven recommendations for packaging would greatly simplify the process and reduce reliance on others. Additionally, AI tools that can alert users to scratches, stains, or other imperfections could assist visually impaired users in capturing accurate product descriptions and minimizing post-sale disputes. This level of support would greatly enhance their independence and confidence in the packaging and shipping stages of the selling process. Furthermore, integrating digital shipping tools that allow users to input shipping details and print labels at home could significantly reduce reliance on others. While participants like P3 and P5 expressed an interest in these services, they were often unaware of how to access them. Implementing these tools within the C2C platform itself, with clear instructions and voice-guided options, could empower users to complete shipping tasks without assistance, addressing one of the major offline barriers identified in the study.

Platform Usability and Accessible Interface Design. Improving the usability of C2C platforms is crucial for enabling visually impaired users to manage the selling process independently. Many participants, such as P1 and P7, expressed frustration with cluttered interfaces, small text, and poor integration with screen readers. These issues significantly hindered their ability to list items and interact with buyers. For example, P1 commented, *“the app feels cramped, and the images are too small to properly see.”* To address these issues, C2C platforms should adopt inclusive design principles prioritising accessibility for visually impaired users. This could include features such as (1) Customizable text sizes that do not interfere with the platform layout, (2) Voice-activated commands for critical tasks such as listing items, managing buyer communication, and tracking shipments, and (3) High-contrast mode options that ensure platform elements are easily distinguishable. Incorporating these features into the platform’s core design would empower visually impaired users to engage with the selling process without needing assistance. Moreover, screen reader compatibility should be seamless, with user interfaces optimized for the specific needs of those who rely on such technologies.

Assistive Technologies for Item Photography. Taking clear and accurate photos of items is a crucial part of the C2C selling process but

presents significant challenges for visually impaired users. Assistive technologies that guide users in framing and positioning items correctly for photographs could significantly reduce frustration at this stage of the selling process. P4, for example, struggled with aligning items in the frame and ensuring they were well-captured: *I struggle to get the angle right in photos; I can't tell if it's framed properly.* To mitigate this, AI-based photo tools could provide real-time guidance during the photography process, using audio or haptic feedback to help users centre and align items. These tools could also suggest lighting, focus, and positioning adjustments, ensuring that visually impaired users can capture high-quality images that accurately represent the items they are selling. Additionally, automatic item detection and description generation could further assist users in ensuring that product details are conveyed effectively to potential buyers. These technologies would enhance the visual representation of items and reduce user reliance on family or friends for assistance, as many participants reported needing help for such tasks.

Integrated Post-Sale Management and Feedback Tracking. Another area that emerged from the study was the difficulty visually impaired users faced in tracking shipments and managing post-sale interactions, such as responding to feedback or handling buyer queries. For participants like P4 and P6, navigating between multiple apps to check delivery statuses or read buyer reviews was cumbersome, and the small text used in these interfaces made it even more difficult to manage these tasks independently. P6 noted, *I cannot see, so I'm not interested in seeing tracking status.* A potential solution to this issue is the development of integrated post-sale management systems that consolidate all necessary functions—tracking, buyer communication and feedback—into one app or platform. This system could be enhanced with notifications and alerts sent through accessible messaging platforms such as LINE⁵ or WhatsApp⁶, which participants are more comfortable using. Additionally, these systems should include voice-activated feedback tools that allow users to read, categorize, and respond to buyer comments without relying on sighted assistance. By streamlining post-sale tasks and making them accessible, these innovations could significantly improve visually impaired users' engagement with buyers and allow them to maintain control over the entire selling process.

6 Conclusion

This study explored the accessibility challenges visually impaired users face on C2C platforms, specifically focusing on the selling process. Our findings revealed significant barriers across multiple stages, from item listing and packaging to post-sale interactions, primarily due to inaccessible platform interfaces and the reliance on external assistance. The study also highlighted key opportunities for improvement through the implementation of assistive technologies, such as AI-powered packaging tools, accessible photo-taking assistance, and integrated post-sale management systems. By addressing these challenges with inclusive design innovations, C2C platforms can empower visually impaired users to engage independently in the selling process, fostering greater autonomy and satisfaction.

⁵<https://line.me/en/>

⁶<https://www.whatsapp.com/>

References

- [1] Natã M. Barbosa, Jordan Hayes, Smirity Kaushik, and Yang Wang. 2022. "Every Website Is a Puzzle!": Facilitating Access to Common Website Features for People with Visual Impairments. *ACM Trans. Access. Comput.* 15, 3, Article 19 (July 2022), 35 pages. <https://doi.org/10.1145/3519032>
- [2] Jeffrey P. Bigham, Ryan S. Kaminsky, Richard E. Ladner, Oscar M. Danielsson, and Gordon L. Hempton. 2006. WebInSight: making web images accessible. In *Proceedings of the 8th International ACM SIGACCESS Conference on Computers and Accessibility* (Portland, Oregon, USA) (ASSETS '06). Association for Computing Machinery, New York, NY, USA, 181–188. <https://doi.org/10.1145/1168987.1169018>
- [3] Shih-Chieh Dai, Aiping Xiong, and Lun-Wei Ku. 2023. LLM-in-the-loop: Leveraging large language model for thematic analysis. *arXiv preprint arXiv:2310.15100* (2023).
- [4] Maitraye Das, Alexander J. Fiannaca, Meredith Ringel Morris, Shaun K. Kane, and Cynthia L. Bennett. 2024. From Provenance to Aberrations: Image Creator and Screen Reader User Perspectives on Alt Text for AI-Generated Images. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '24). Association for Computing Machinery, New York, NY, USA, Article 900, 21 pages. <https://doi.org/10.1145/3613904.3642325>
- [5] Zhitong Guan, Zeyu Xiong, and Mingming Fan. 2024. FetchAid: Making Parcel Lockers More Accessible to Blind and Low Vision People With Deep-learning Enhanced Touchscreen Guidance, Error-Recovery Mechanism, and AR-based Search Support. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '24). Association for Computing Machinery, New York, NY, USA, Article 39, 15 pages. <https://doi.org/10.1145/3613904.3642213>
- [6] Simon Harper and Sean Bechhofer. 2007. SADLE: Structural semantics for accessibility and device independence. *ACM Trans. Comput.-Hum. Interact.* 14, 2 (Aug. 2007), 10–es. <https://doi.org/10.1145/1275511.1275516>
- [7] Naoki Hirabayashi, Masakazu Iwamura, Zheng Cheng, Kazunori Minatani, and Koichi Kise. 2023. VisPhoto: Photography for People with Visual Impairments via Post-Production of Omnidirectional Camera Imaging. In *Proceedings of the 25th International ACM SIGACCESS Conference on Computers and Accessibility* (New York, NY, USA) (ASSETS '23). Association for Computing Machinery, New York, NY, USA, Article 6, 17 pages. <https://doi.org/10.1145/3597638.3608422>
- [8] Mina Huh, Yi-Hao Peng, and Amy Pavel. 2023. GenAssist: Making Image Generation Accessible. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology* (San Francisco, CA, USA) (UIST '23). Association for Computing Machinery, New York, NY, USA, Article 38, 17 pages. <https://doi.org/10.1145/3586183.3606735>
- [9] Yuka Kaniwa, Masaki Kuribayashi, Seita Kayukawa, Daisuke Sato, Hironobu Takagi, Chieko Asakawa, and Shigeo Morishima. 2024. ChitChatGuide: Conversational Interaction Using Large Language Models for Assisting People with Visual Impairments to Explore a Shopping Mall. *Proc. ACM Hum.-Comput. Interact.* 8, MCHI, Article 247 (Sept. 2024), 25 pages. <https://doi.org/10.1145/3676492>
- [10] Patrick E. Lanigan, Aaron M. Paulos, Andrew W. Williams, Dan Rossi, and Priya Narasimhan. 2006. Trinetra: Assistive Technologies for Grocery Shopping for the Blind.. In *Proceedings of the 10th IEEE International Symposium on Wearable Computers (ISWC)*. IEEE, 147–148.
- [11] Dan Lin and Rui Zhang. 2015. Research on the impact mechanism between logistics service innovation and online customer repurchase intention. In *2015 International Conference on Management Science and Management Innovation (MSMI 2015)*. Atlantis Press, 251–260.
- [12] Guanhong Liu, Xianghua Ding, Chun Yu, Lan Gao, Xingyu Chi, and Yuanchun Shi. 2019. "I Bought This for Me to Look More Ordinary": A Study of Blind People Doing Online Shopping. In *Proc. of the 2019 CHI Conf. on Human Factors in Computing Systems (CHI '19)*. Assoc. for Computing Machinery, New York, 1–11. <https://doi.org/10.1145/3290605.3300602>
- [13] Diego López-de Ipiña, Tania Lorido, and Unai López. 2011. Blindshopping: enabling accessible shopping for visually impaired people through mobile technologies. In *Proceedings of the 9th International Conference on Toward Useful Services for Elderly and People with Disabilities: Smart Homes and Health Telematics* (Montreal, Canada) (ICOST'11). Springer-Verlag, Berlin, Heidelberg, 266–270.
- [14] Yoshiki Nagatani, Kazuki Takazawa, Masaaki Sadasue, Mariko Kobayashi, Kentaro Teramoto, Nozomu Yoshida, Kotaro Oomori, Akihisa Shitara, Koyomi Kobayashi, Aiko Kuroda, and Misato Takahashi. 2022. How Visually Impaired People Use Information Media and E-Commerce in Japan. In *HCI International 2022 Posters*, Constantine Stephanidis, Margherita Antona, and Stavroula Ntoa (Eds.). Springer International Publishing, Cham, 553–561.
- [15] John Nicholson, Vladimir Kulyukin, and Daniel Coster. 2009. ShopTalk: independent blind shopping through verbal route directions and barcode scans. *The Open Rehabilitation Journal* 2, 1 (2009), 11–23.
- [16] OpenAI. 2024. GPT-4o. Retrieved October 10 2024 from <https://platform.openai.com/docs/models/gpt-4o>.
- [17] Yash Prakash, Mohan Sunkara, Hae-Na Lee, Sampath Jayarathna, and Vikas Ashok. 2023. AutoDesc: Facilitating Convenient Perusal of Web Data Items for Blind Users. In *Proceedings of the 28th International Conference on Intelligent*

- User Interfaces* (Sydney, NSW, Australia) (*IUI '23*). Association for Computing Machinery, New York, NY, USA, 32–45. <https://doi.org/10.1145/3581641.3584049>
- [18] Bektur Ryskeldiev, Kotaro Hara, Mariko Kobayashi, and Koki Kusano. 2022. Investigating Accessibility Challenges and Opportunities for Users with Low Vision Disabilities in Customer-to-Customer (C2C) Marketplaces. In *Proceedings of the 24th International ACM SIGACCESS Conference on Computers and Accessibility* (Athens, Greece) (*ASSETS '22*). Association for Computing Machinery, New York, NY, USA, Article 79, 4 pages. <https://doi.org/10.1145/3517428.3550390>
- [19] Abigale Stangl, Meredith Ringel Morris, and Danna Gurari. 2020. "Person, Shoes, Tree. Is the Person Naked?" What People with Vision Impairments Want in Image Descriptions. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (*CHI '20*). Association for Computing Machinery, New York, NY, USA, 1–13. <https://doi.org/10.1145/3313831.3376404>
- [20] Abigale J. Stangl, Esha Kothari, Suyog D. Jain, Tom Yeh, Kristen Grauman, and Danna Gurari. 2018. BrowseWithMe: An Online Clothes Shopping Assistant for People with Visual Impairments. In *Proceedings of the 20th International ACM SIGACCESS Conference on Computers and Accessibility* (Galway, Ireland) (*ASSETS '18*). Association for Computing Machinery, New York, NY, USA, 107–118. <https://doi.org/10.1145/3234695.3236337>
- [21] Sarit Szpiro, Yuhang Zhao, and Shiri Azenkot. 2016. Finding a store, searching for a product: a study of daily challenges of low vision people. In *Proceedings of the 2016 ACM International Joint Conference on Pervasive and Ubiquitous Computing* (Heidelberg, Germany) (*UbiComp '16*). Association for Computing Machinery, New York, NY, USA, 61–72. <https://doi.org/10.1145/2971648.2971723>
- [22] Alexandra Vtyurina, Adam Fourney, Meredith Ringel Morris, Leah Findlater, and Ryan W. White. 2019. VERSE: Bridging Screen Readers and Voice Assistants for Enhanced Eyes-Free Web Search. In *Proceedings of the 21st International ACM SIGACCESS Conference on Computers and Accessibility* (Pittsburgh, PA, USA) (*ASSETS '19*). Association for Computing Machinery, New York, NY, USA, 414–426. <https://doi.org/10.1145/3308561.3353773>
- [23] Ruolin Wang, Zixuan Chen, Mingrui Ray Zhang, Zhaoheng Li, Zhixiu Liu, Zihan Dang, Chun Yu, and Xiang 'Anthony' Chen. 2021. Revamp: Enhancing Accessible Information Seeking Experience of Online Shopping for Blind or Low Vision Users. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems* (Yokohama, Japan) (*CHI '21*). Association for Computing Machinery, New York, NY, USA, Article 494, 14 pages. <https://doi.org/10.1145/3411764.3445547>
- [24] Yanan Wang, Ruobin Wang, Crescentia Jung, and Yea-Seul Kim. 2022. What makes web data tables accessible? Insights and a tool for rendering accessible tables for people with visual impairments. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems* (New Orleans, LA, USA) (*CHI '22*). Association for Computing Machinery, New York, NY, USA, Article 595, 20 pages. <https://doi.org/10.1145/3491102.3517469>
- [25] Michele A. Williams, Amy Hurst, and Shaun K. Kane. 2013. "Pray before you step out": describing personal and situational blind navigation behaviors. In *Proceedings of the 15th International ACM SIGACCESS Conference on Computers and Accessibility* (Bellevue, Washington) (*ASSETS '13*). Association for Computing Machinery, New York, NY, USA, Article 28, 8 pages. <https://doi.org/10.1145/2513383.2513449>
- [26] Nancy Worth. 2013. Visual impairment in the city: Young people's social strategies for independent mobility. *Urban Studies* 50, 3 (2013), 574–586.